

Mohammad Savari, EIT

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Academic Profile

Ph.D. candidate in Mechanical Engineering at the University of Ottawa with research expertise in complex systems, social contagion, consensus formation, information propagation, control systems, network science, and computational modeling. Current research develops modeling frameworks to understand how social contagion mechanisms shape collective consensus in complex networks. Additional research experience includes complex global networks, inter-organizational alliance networks, community detection, web scraping, NLP-assisted data collection, and network visualization using Python and Gephi.

Research Interests

Complex systems; network science; social contagion; consensus formation; collective decision-making; information propagation; control and system dynamics; community detection; multi-agent systems; reinforcement learning; computational social science; organizational networks.

Education

Ph.D. in Mechanical Engineering <i>University of Ottawa</i>	2021–2026 expected <i>Ottawa, Canada</i>
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- Thesis: *Fostering Consensus Using Social Contagion*.
- Conduct doctoral research in complex systems, social contagion, consensus formation, and computational network modeling.
- Granted direct admission to the Ph.D. program from a bachelor's degree.
- Awarded a full admission scholarship and merit scholarship from the University of Ottawa.

Bachelor of Science in Electrical Engineering – Control Systems <i>Arak University of Technology</i>	2014–2018 <i>Arak, Iran</i>
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Research Experience

Complex Networks Researcher <i>Telfer School of Management, University of Ottawa</i>	May 2025–Present <i>Ottawa, Canada</i>
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- Identify and characterize sub-networks, communities, and topological properties in complex global networks using network science methodologies.
- Collect, clean, and integrate data through web scraping, natural language processing, manual extraction, and entity-matching protocols.
- Investigate mechanisms driving network formation, network evolution, and inter-organizational relationships within complex systems.
- Generate network visualizations, computational models, and exploratory analyses using Python and Gephi.
- Examine collaboration patterns and classify organizational roles, including NGOs, multinational corporations, and non-state actors, within interconnected alliance networks.

Social Contagion Researcher <i>Faculty of Engineering, University of Ottawa</i>	Sep 2021–Present <i>Ottawa, Canada</i>
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- Develop computational models to study social contagion, consensus formation, and information propagation on complex networks.

- Contribute to a new modeling framework that integrates system dynamics into the analysis of social contagion processes.
- Evaluate existing measures of information propagation and assess their applicability to dynamic networked systems.
- Propose system dynamics as a key property for improving the analysis of network behavior and collective decision-making.
- Develop heterogeneous scale-free network models with tunable clustering properties.
- Implement distribution likelihood analyses to evaluate scale-free characteristics in generated network models.
- Prepare research documentation, technical reports, presentations, and manuscript materials.

Teaching Experience

Teaching Assistant

University of Ottawa

Jan 2022–Present

Ottawa, Canada

- Support undergraduate and graduate-level courses through tutorials, laboratories, grading, student support, and course administration.
- Assist students with system dynamics, mechatronics, biomedical systems, robotics, product development, and machine learning for control concepts.
- Provide feedback on assignments, projects, and technical submissions to strengthen student learning outcomes.
- Supported courses: Mechatronics; Systems Dynamics; Biomedical Systems Dynamics; Introduction to Product Development for Engineers and Computer Scientists; Machine Learning in Control; Robotics.

Publications

- **Savari, M.**, et al. 2025. “How Social Contagions Shape Collective Consensus in the Presence of Scale-Free Networks.” Submitted.
- Kwa, H. L., Kit, J. L., Horsevad, N., Philippot, J., **Savari, M.**, & Bouffanais, R. 2023. “Adaptivity: A Path Towards General Swarm Intelligence?” *Frontiers in Robotics and AI*, 10, 1163185.

Conference Presentations

- **Savari, M.**, et al. 2025. “Leveraging Social Contagion to Foster Consensus in Collective Decision-Making.” Abstract paper presented at the International Conference on Computational Science, Singapore.

Technical and Research Skills

Programming and Data Analysis: Python, R, MATLAB, LabVIEW, statistical analysis, data processing, computational modeling, data-driven insights

Network Science and Complex Systems: Complex systems modeling, social contagion analysis, consensus formation, information propagation, community detection, scale-free networks, multi-agent systems, reinforcement learning

Research Methods: Literature review, experimental design, simulation-based research, technical writing, manuscript preparation, conference presentations, research documentation

Tools and Platforms: Python, R, MATLAB, LabVIEW, Gephi, Arduino, Atmel microcontrollers,

Applied Engineering and Professional Experience

Project Manager

Sep 2024–Present

Centre for Entrepreneurship and Engineering Design

Ottawa, Canada

- Manage four interdisciplinary student teams developing accessibility-focused engineering devices from concept to prototype.
- Review technical documentation, design reports, and project deliverables to ensure alignment with client requirements.
- Coordinate procurement of components, materials, and resources required for prototype development and implementation.

System Engineering Intern

Sep 2023–Mar 2024

AirShare Inc.

Ottawa, Canada

- Developed and implemented system models to support performance optimization and design validation.
- Supported sensor calibration, sensor fusion, data validation, field testing, troubleshooting, and technical documentation.
- Established communication frameworks to improve system reliability and operational stability.

Search Engine Evaluator

Jun 2024–Present

TELUS International

- Evaluate digital information quality, relevance, accuracy, completeness, and alignment with user intent.
- Apply structured evaluation guidelines with consistency, attention to detail, and analytical judgment.

Voluntary and Community Service

Mechanical Engineering Departmental Director

Jun 2021–Apr 2025

Graduate Student's Association, University of Ottawa

Ottawa, Canada

- Represented graduate students in the Mechanical Engineering Department.
- Prepared communication materials, responded to student inquiries, and supported administrative coordination.
- Assisted with submission, review, documentation, event coordination, and preparation of official materials.